30 Light of wavelength λ is incident normally on a diffraction grating with a total number of N lines in width w.

A second order maximum is observed at an angle of diffraction θ.

What is N?

A $\frac{w\lambda}{\sin\theta}$ **B** $\frac{2w\lambda}{\sin\theta}$ **C** $\frac{w\sin\theta}{2\lambda}$ **D** $\frac{w\sin\theta}{\lambda}$